



Alberta Public Health Disease Management Guidelines

St. Louis Encephalitis



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Contents

Case Definition	4
Confirmed Case	4
Probable Case	4
Physicians, Health Practitioners and Others	5
Laboratories	5
Alberta Health Services and First Nations and Inuit Health Branch	5
Epidemiology.....	6
Etiology	6
Clinical Presentation.....	6
Diagnosis	6
Treatment.....	6
Reservoir.....	6
Transmission.....	6
Incubation Period	6
Period of Communicability.....	7
Host Susceptibility	7
Incidence.....	7
Public Health Management.....	8
Key Investigation.....	8
Management of a Case.....	8
Management of Contacts	8
Preventive Measures.....	8
Appendix 1: Revision History	9
References.....	10

Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Isolation of St. Louis encephalitis (SLE) virus from an appropriate clinical specimen (e.g., serum, CSF)^(B)
- OR
- Detection of SLE virus antigen in tissue, blood or CSF
- OR
- Detection of SLE viral nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., plasma, CSF, serum)^(B)
- OR
- Seroconversion or significant difference between acute and convalescent phase SLE Hemagglutination Inhibition Assay (HAI) titers ideally taken at least two weeks apart in an appropriate clinical specimen (e.g., serum)^(B) and confirmed by Plaque Reduction Neutralization Assay (PRNT)^(C)

Probable Case

Clinical illness^(A) and one of the following:

- Seroconversion or significant difference between acute and convalescent phase SLE HAI titres ideally taken at least two weeks apart but not confirmed by PRNT^(C)
- OR
- Stable elevated serial HAI titres^(D) to SLE that occur during a period when and where arboviral transmission is likely^(E)
- OR
- Single elevated HAI titre^(D) to SLE that occurs during a period when and where arboviral transmission is likely^(E)

^(A) Clinical illness is characterized by a febrile illness of variable severity associated with neurological symptoms ranging from headache to aseptic meningitis or encephalitis. Arboviral encephalitis cannot be distinguished clinically from other central nervous system (CNS) infections. Symptoms can include headache, confusion or other alteration in sensorium, nausea and vomiting. Signs may include fever, meningismus, cranial nerve palsies, paresis or paralysis, sensory deficits, altered reflexes, convulsions, abnormal movements, and coma of varying degree.

^(B) Refer to the [National Microbiology Laboratory \(NML\) Guide to Services](#) for current specimen collection and submission information.

^(C) Seroconversion indicates recent infection with a flavivirus (e.g., dengue fever, California Serogroup viruses or West Nile virus) but cannot pinpoint which one due to antibody cross-reactivity. These are considered probable cases. PRNT confirmatory testing may be requested at NML through the Public Health Laboratory (ProvLab).

^(D) A single elevated HAI titre of $\geq 1:320$ is suggestive of recent infection. It is recommended that a second specimen be collected. A second laboratory result with a stable (unchanged/static or $\leq$ twofold rise) elevated HAI titre is also suggestive of recent infection.

^(E) SLE exposure is most likely to occur during the summer and early fall in most of the United States (US) and Canada and year round in Central America and Brazil.⁽¹⁾

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Reference 2 applies to this section.

SLE is a member of the Flaviviridae *flavivirus* family. SLE is susceptible to many disinfectants and is inactivated by heat. As well it is sensitive to UV and gamma irradiation. It is stable in aerosol form for up to six hours at room temperature.⁽³⁾

Clinical Presentation

Reference 2 applies to this section.

SLE is the leading cause of epidemic flaviviral encephalitis in the US, even though less than 1% of infections are clinically apparent and the majority of infections remain undiagnosed. Early symptoms include headache, fever, myalgia, malaise, nausea and/or vomiting, photophobia, sore throat, dysuria, and urinary frequency. Severe infection is characterized by acute onset, with headache, high fever, nausea, myalgia, and malaise followed by meningeal signs, stupor, coma, convulsions and paralysis. The disease is generally milder in children than in adults, and the elderly are at highest risk for acute encephalitis and death. Neurologic sequelae occur in about 10% of SLE cases and 30–50% of severe cases have prolonged convalescence. The case fatality rate in the US is 3–38% (especially in the elderly).

Diagnosis

Diagnosis of SLE is made through rapid serologic assays such as IgM capture ELISA and IgG ELISA. In early infection IgM antibody is more specific, while later in infection IgG antibody is more reactive. Viral isolation and identification have been useful in defining viral agents in serum, CSF and mosquito vectors.

Treatment

- Antibiotics are not effective.
- No effective antiviral treatment has been identified.

Reservoir

The reservoir for SLE is most likely birds. The virus may overwinter in birds, rodents, bats, reptiles, amphibians or surviving mosquito eggs/adult mosquitos. The host range includes human, wild birds, and other mammals.

Transmission

SLE is the most common mosquito-transmitted human pathogen in the US. It is maintained in the summer season by a bird-mosquito cycle. Mosquitoes feed on wild birds and occasionally pass the virus on to humans. Vectors are members of the *Culex* spp. Transmission occurs most often through the bite of a *Culex nigrapalpus* mosquito.⁽⁴⁾ *Culex pipiens*, *C. tarsalis*, *C. quinquefasciatus* and *C. nigripalpus* are other known vectors.

Incubation Period

The incubation period of SLE is usually four to 21 days.

Period of Communicability

SLE is not directly transmitted from person-to-person. The virus is not demonstrable in blood or CSF after onset of disease. In birds, the viremia lasts two to five days. Mosquitoes are infected for life (one to four months) and vertical transmission (female to egg) may exist. Viremia in horses is rarely present in high titres for long periods. In the case of SLE, viremia typically lasts seven to 10 days.

Host Susceptibility

The elderly, individuals living in low-income areas, and persons with HIV infection are at greater risk of SLE than the general population.

Incidence

General

The virus that causes SLE is present in 48 US states and extends northward into Canada and southward into Central and South America. An outbreak of SLE/West Nile-like virus activity occurred in New York State in 1999. SLE occurs in Northern, Southern, and Central America. Cases occur most often in temperate altitudes in summer and early fall and are commonly limited to areas and years of high temperature and abundant mosquito populations.⁽⁵⁾ Epidemics of SLE recur at irregular intervals (about every 10 to 20 years).

Canada

SLE is not thought to be endemic to Canada but periodically crosses the border as an extension of activity in the central and western US.⁽⁶⁾

Alberta

There have been no cases of SLE reported in Alberta between 1994 and 2020.

Public Health Management

Key Investigation

- Identify the potential source of the infection based on:
 - history of travel to endemic areas,
 - mode of transmission, and
 - incubation period.
- Assess for history of mosquito bites.

Management of a Case

- Supportive care, including IV fluids and ventilation.
- Symptomatic treatment.

Management of Contacts

The infection is not passed from person-to-person.

Preventive Measures

- Educate the public about personal protective measures including:
 - reducing time outdoors at dusk,
 - wearing long pants and long-sleeved shirts, and
 - applying mosquito repellent containing DEET.
- Spray insecticides to kill larvae and adult mosquitoes.
- Support surveillance programs to allow for rapid identification of dangerous viruses in mosquito populations; this can shorten public health response time and reduce the spread of infected vectors.
- Laboratory safety – Arboviruses require biosafety level 3 practices.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology • Key Investigation section moved to Public Health Management (formerly called Control) • Updated web links
	Incidence	• Updated Alberta Incidence

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